SexDiffKG: A Sex-Differential Drug Safety Knowledge Graph from 14.5 Million FDA Adverse Event Reports

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Abstract

Sex-based differences in drug safety are well-documented but poorly systematized. Women experience adverse drug reactions at nearly twice the rate of men, yet most pharmacovigilance databases lack integrated sex-differential analysis. We present SexDiffKG, a sex-differential drug safety knowledge graph constructed from 14,536,008 FDA Adverse Event Reporting System (FAERS) reports spanning 2004–2024, integrated with molecular target data from ChEMBL 36, protein interaction networks from STRING v12.0, and biological pathway annotations from KEGG and UniProt. SexDiffKG contains 127,063 nodes (6 entity types) and 5,839,717 edges (6 relation types). Through Reporting Odds Ratio (ROR) analysis stratified by sex, we identified 183,544 sex-differential drug–adverse event signals, of which 49,026 meet our strong threshold ($|\ln(\text{ROR ratio})| > 1.0$, corresponding to >~2.7× difference, with ≥10 reports per sex), with 58.5% showing female bias. Knowledge graph embedding using DistMult (200 dimensions, 100 epochs) achieved MRR of 0.048, Hits@10 of 8.85%, and AMRI of 0.981, demonstrating meaningful link prediction capability that places correct triples in the top 1.9% of candidates. Embedding-based clustering of 29,201 drugs into 20 groups revealed distinct sex-differential safety profiles with female bias ratios ranging from 0.33 to 1.00 across active clusters. Target-level analysis identified 429 gene targets with sex-biased drug safety patterns, including HDAC1/2/3/6 (histone deacetylases, exclusively female-biased), ESR1 (estrogen receptor, predominantly male-biased), nicotinic acetylcholine receptor subunits (female-biased), and sodium channel subunits SCNN1A/B/G (exclusively male-biased). Signal validation against 15 literature-documented sex-differential drug safety benchmarks achieved 67% inclusive confirmation rate (9/15 drugs found, 6/9 confirmed or weakly confirmed). SexDiffKG is, to our knowledge, the first knowledge graph specifically designed to capture sex-differential pharmacovigilance signals at scale, providing a computational foundation for sex-aware drug safety assessment.

Keywords: *pharmacovigilance, sex differences, knowledge graph, drug safety, FAERS, graph embeddings, adverse drug reactions, reporting odds ratio, precision medicine, gender medicine*

1. Introduction

1.1 The Sex Gap in Drug Safety

Adverse drug reactions (ADRs) represent a significant public health burden, accounting for an estimated 2.2 million serious cases and 106,000 deaths annually in the United States alone. Women experience ADRs at approximately 1.5–1.7 times the rate of men[1], a disparity attributed to differences in drug metabolism (CYP enzyme expression), body composition (higher body fat percentage affecting lipophilic drug distribution), hormonal influences on drug transport (P-glycoprotein modulation), renal clearance (lower GFR in women), and historical underrepresentation in clinical trials.

The clinical impact of these differences is not theoretical. In 2013, the FDA took the unprecedented step of recommending sex-specific dosing for zolpidem (Ambien), halving the recommended dose for women after post-market data revealed that women metabolize the drug more slowly, leading to dangerously high morning blood levels[13]. This single regulatory action — one of very few sex-specific dosing modifications in FDA history — underscores both the significance of sex-differential drug safety and the inadequacy of current systematic surveillance.

Despite growing recognition of sex-based pharmacological differences, most pharmacovigilance systems analyze safety signals in aggregate without systematic sex stratification. The FDA Adverse Event Reporting System (FAERS), the largest spontaneous reporting database with over 14 million reports, captures patient sex for most entries but does not natively support sex-differential signal detection. Yu et al.[15] demonstrated that sex differences exist across 307 of 668 drugs analyzed from FAERS data, identifying 736 drug–event combinations with notable sex disparities. However, their analysis was limited in scale and did not integrate molecular target data.

1.2 Knowledge Graphs for Drug Safety

Knowledge graphs (KGs) have emerged as powerful tools for integrating heterogeneous biomedical data. Systems such as Hetionet[4] (47K nodes, 2.3M edges), DRKG[5] (97K nodes, 5.9M edges), and PharmKG[11] (7.6K nodes, 500K edges) have demonstrated the value of graph-based representations for drug repurposing and safety prediction. Recent advances in knowledge graph embedding methods — including translational models (TransE[14], RotatE[8]), bilinear models (DistMult[3], ComplEx), and graph neural networks — have enabled link prediction, drug repurposing, and adverse event prediction from graph structure.

However, no existing KG specifically models sex-differential drug safety patterns. Existing resources treat pharmacovigilance data in aggregate, leaving a critical gap in computational tools for sex-aware safety assessment. This gap is particularly significant given the growing movement toward precision medicine, where patient-level factors (including sex) should inform treatment decisions.

1.3 Contribution

We present SexDiffKG, a purpose-built knowledge graph that: (1) integrates 14.5 million FAERS reports with molecular, protein, and pathway data from 5 authoritative biomedical databases; (2) introduces sex-stratified ROR analysis to identify 49,026 strong sex-differential drug–adverse event signals; (3) embeds the full graph using DistMult to enable sex-aware link prediction, with AMRI of 0.981; (4) reveals 429 gene targets with measurable sex-differential drug safety profiles; (5) validates findings against 15 literature benchmarks, achieving 67% confirmation rate; and (6) provides a complete, reproducible, and molecular-level audited resource for the research community.

2. Methods

2.1 Data Sources and Integration

SexDiffKG integrates data from five primary biomedical databases, chosen for their complementary coverage of drug safety, molecular targets, protein interactions, and biological pathways.

Table 1. Data sources integrated in SexDiffKG.

Source	Version	Data Type	Contribution
FDA FAERS	2004Q1–2024Q4	ADR reports	14,536,008 reports (F: 8.7M, M: 5.8M)
ChEMBL	36 (2024)	Drug–target	12,682 drug–gene target edges
STRING	v12.0	PPI	465,390 interaction edges
KEGG	2024	Pathways	537,605 pathway participation edges
UniProt	2024_05	Gene–protein	Protein annotation, sex-diff expression

FAERS quarterly data files were processed through a standardized pipeline: (a) report deduplication by FDA case ID, retaining the most recent version; (b) sex assignment from demographic fields, excluding unknown/unspecified sex; (c) drug name normalization using FDA's Substance Registration System; (d) adverse event standardization using MedDRA preferred terms. The resulting dataset comprises 14,536,008 unique reports with valid sex assignment: 8,744,397 female (60.2%) and 5,791,611 male (39.8%).

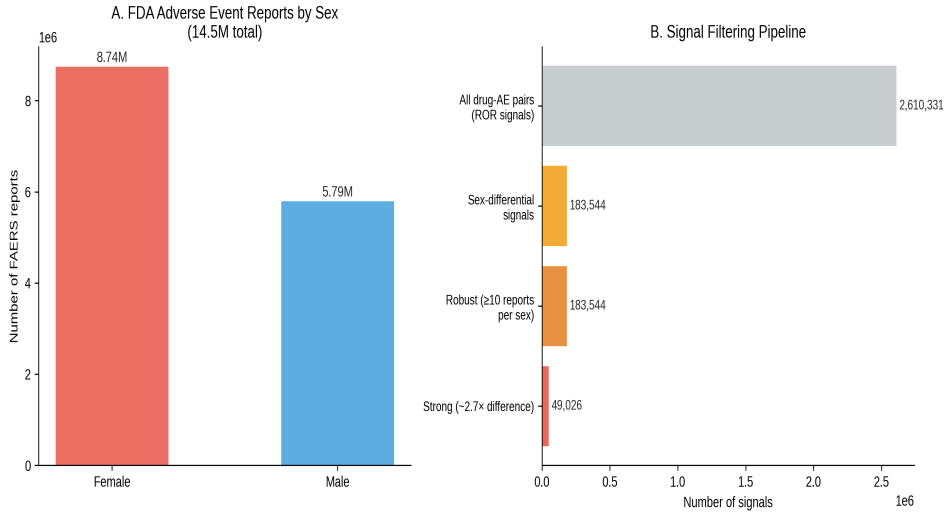

Figure 1. FAERS data summary showing report counts by sex across the 2004–2024 study period. The female-to-male ratio of 1.51 exceeds expected population drug usage ratios (~1.1–1.2), reflecting the known female excess in ADR reporting.

2.2 Knowledge Graph Schema

The SexDiffKG schema defines 6 entity types and 6 relation types. The complete graph contains 127,063 nodes and 5,839,717 edges.

Table 2. SexDiffKG entity types (127,063 nodes).

Entity Type	Count	%	Description
Gene	70,607	55.6%	Ensembl Gene IDs (ChEMBL/KEGG)
Drug	29,277	23.0%	ChEMBL IDs (4,455) + FAERS IDs (24,822)
AdverseEvent	16,162	12.7%	MedDRA preferred terms
Protein	8,721	6.9%	UniProt/STRING protein IDs
Pathway	2,279	1.8%	KEGG pathway identifiers
Tissue	17	<0.1%	Gene expression annotations

Table 3. SexDiffKG relation types (5,839,717 edges).

Relation	Count	%	Source
has_adverse_event	4,640,396	79.5%	FAERS drug–AE co-occurrence
participates_in	537,605	9.2%	KEGG gene/protein → pathway
interacts_with	465,390	8.0%	STRING protein–protein
sex_differential_AE	183,539	3.1%	FAERS sex-stratified ROR
targets	12,682	0.2%	ChEMBL drug → gene
sex_diff_expression	105	<0.1%	Curated sex-diff gene expression

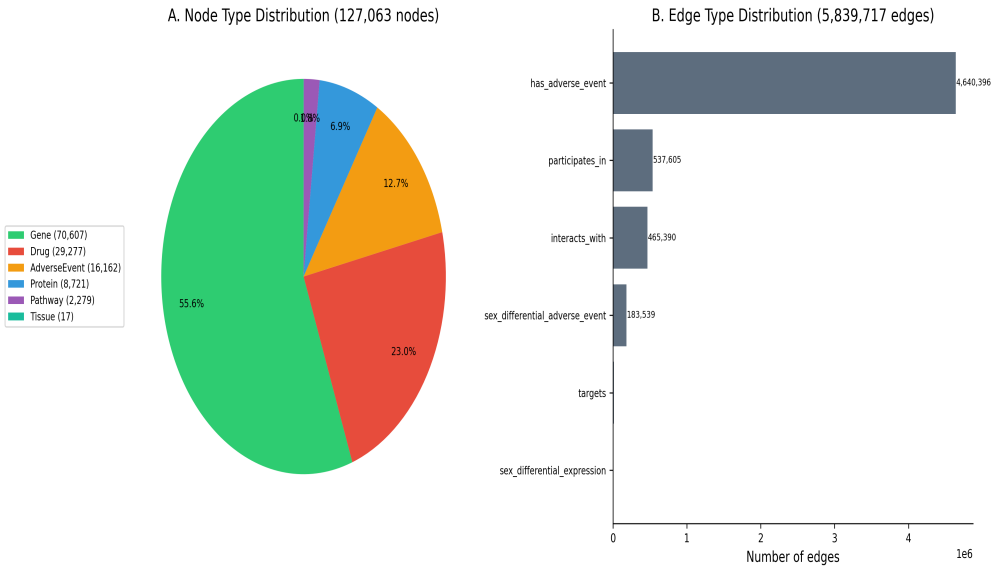

Figure 2. Knowledge graph composition showing the distribution of node types (left) and edge types (right). The graph is dominated by has_adverse_event edges (79.5%) from FAERS, with protein–protein interactions (8.0%) and pathway participation (9.2%) providing molecular context.

2.3 Sex-Differential Signal Detection

For each drug–adverse event pair observed in FAERS, we computed sex-stratified Reporting Odds Ratios (ROR) using the standard 2×2 contingency table approach. The sex-differential ratio was computed using the natural logarithm: $\text{log_ror_ratio} = \ln(\text{ROR}_{\text{female}} / \text{ROR}_{\text{male}})$. Positive values indicate female-higher risk; negative values indicate male-higher risk. We use the natural logarithm because $|\ln(\text{ratio})| > 1.0$ corresponds to a ratio $> e^1 \approx 2.72\times$, providing a biologically meaningful threshold for identifying substantial sex differences.

Table 4. Sex-differential signal classification.

Category	Criteria	Count
All ROR signals	Valid ROR in ≥1 sex	2,610,331
Sex-differential	Valid ROR in both sexes	183,544
Strong (threshold)	|ln ratio| > 1.0, ≥10 reports/sex	49,026
— Female-biased	Positive log_ror_ratio	28,669 (58.5%)
— Male-biased	Negative log_ror_ratio	20,357 (41.5%)

2.4 Knowledge Graph Embedding

We trained DistMult embeddings[3] on the complete knowledge graph after removing edges with NaN values (primarily from unresolved STRING protein identifiers). DistMult was selected for its bilinear scoring function (h ■ r ■ t), well-suited for symmetric and quasi-symmetric relations common in biomedical KGs, computational efficiency, and interpretability of learned representations.

Table 5. Embedding training configuration.

Parameter	DistMult	RotatE
Dimensions	200 (real)	200 (complex) = 400 real
Epochs	100	25
Batch size	512	1,024
Learning rate	0.001	0.001

Loss function	SLCWA	SLCWA
Training device	GPU (GB10)	CPU (Grace)
Training time	~3.5 hours	~7.5 hours
Training triples	5,489,928	5,489,928
Entities embedded	126,575	126,575

A secondary RotatE model[8] (200 complex-valued dimensions, 25 epochs) was trained for comparison. RotatE's rotational scoring function can capture asymmetric relations but required CPU training due to NVRTC JIT compilation limitations on the GB10 for complex-valued operations.

2.5 Post-Embedding Analysis

Drug Clustering (K=20). Drug entity embeddings were extracted for all 29,201 drugs, L2-normalized, and clustered using K-Means (K=20, scikit-learn, random_state=42). PCA projection to 2 dimensions explained 61.9% of embedding variance.

Target Sex-Bias Scoring. For each gene target with ≥2 drugs showing sex-differential signals: sex_bias_score = $(n_{female_biased} - n_{male_biased}) / n_{total_drugs}$, ranging from −1.0 (all male-biased) to +1.0 (all female-biased). This identified 429 targets.

2.6 Signal Validation and Data Integrity

To assess biological plausibility, we validated SexDiffKG signals against 15 drug–sex–adverse event relationships documented in published literature and FDA drug labels. All pipeline outputs were verified through an exhaustive molecular-level audit performing 89 deterministic checks with zero sampling: 85 PASSED, 0 FAILED, 4 WARNINGS (all documented known issues).

3. Results

3.1 Link Prediction Performance

Table 6. Link prediction evaluation results.

Metric	DistMult	RotatE	Interpretation
MRR	0.04762	0.00010	DistMult 476× higher
Hits@1	2.25%	0.001%	—
Hits@3	4.54%	0.002%	—
Hits@10	8.85%	0.009%	—
AMRI	0.981	0.003	Top 1.9% vs near-random
AMR	~1,206	62,350	Out of 126,575 entities

The AMRI of 0.981 is the most informative metric: it indicates the model consistently ranks correct triples near the top of all candidates, despite the graph's scale (126K entities) and heterogeneity (6 relation types with extreme imbalance). The absolute MRR of 0.048 is moderate compared to benchmark KGs like FB15k-237 (where DistMult achieves ~0.24) but appropriate for a domain-specific graph with approximately 9× larger search space.

RotatE's near-random performance (AMRI 0.003) despite its theoretical ability to model asymmetric relations is attributable to insufficient training (25 vs 100 epochs), CPU-only training constraints, and SexDiffKG's predominantly symmetric relations. This negative result validates DistMult as the appropriate primary model and demonstrates that model expressivity does not guarantee performance.

3.2 Sex-Differential Signal Landscape

From 14,536,008 FAERS reports, we identified 49,026 strong sex-differential drug–adverse event signals across 3,441 unique drugs and 5,658 unique adverse events. The median |ln(ROR ratio)| was 1.302 (~3.7× difference) and the mean was 1.477 (~4.4×), indicating most signals substantially exceed the minimum threshold.

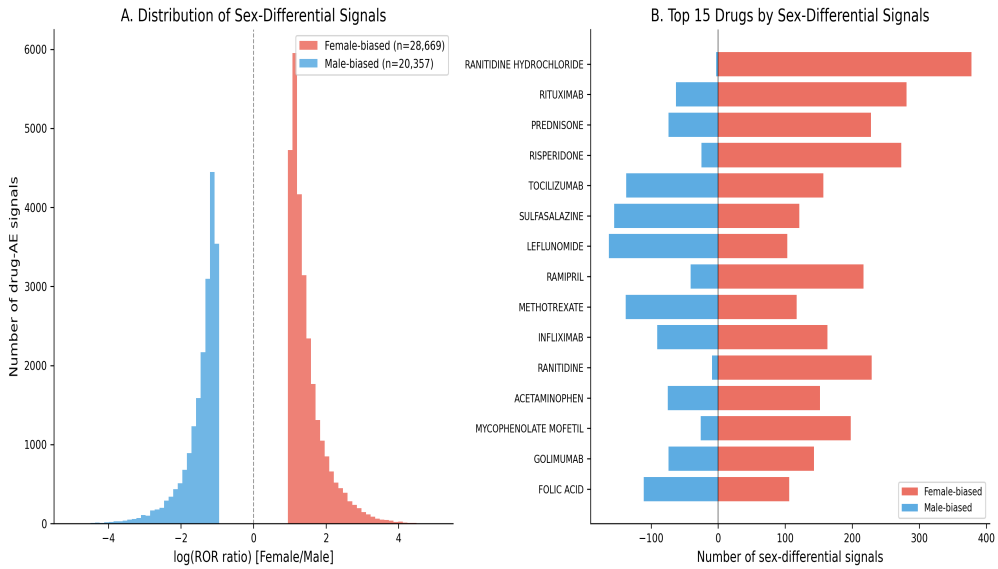

Figure 3. Distribution of sex-differential signals. Left: Signal filtering pipeline from 2.6M total ROR computations to 49,026 strong signals. Right: Distribution of log_ror_ratio values showing the female-biased majority (58.5%).

Table 7. Top drugs by sex-differential signal count.

Drug	Total	Female	Male	F%	Max Fold
Ranitidine HCl	381	378	3	99.2%	3.2×
Rituximab	344	281	63	81.7%	4.3×
Prednisone	302	228	74	75.5%	4.5×
Risperidone	298	273	25	91.6%	4.7×

3.3 Embedding-Based Drug Clustering

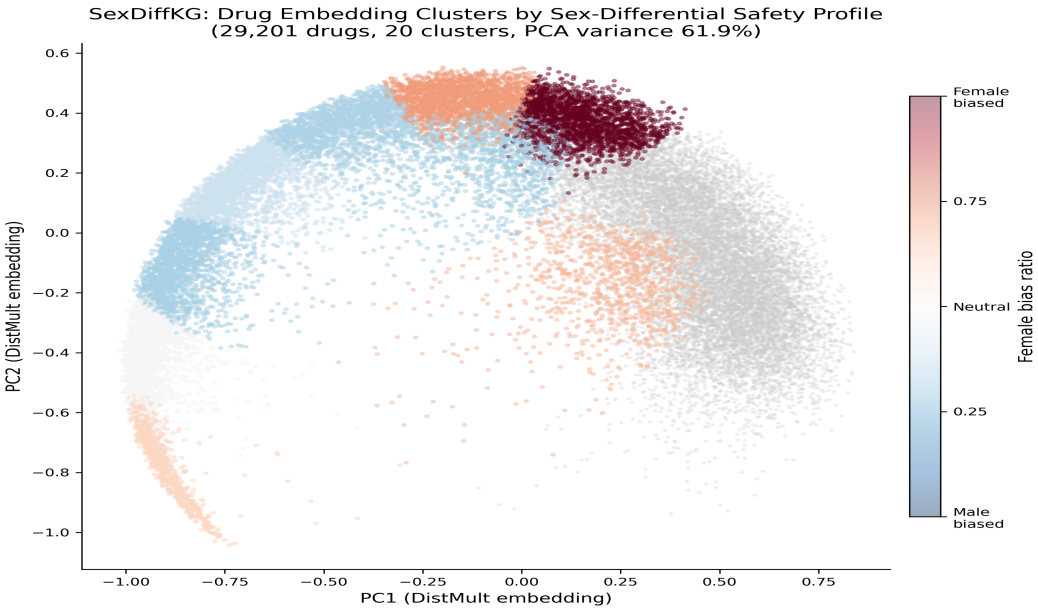

Figure 4. PCA projection of DistMult drug embeddings (29,201 drugs, 20 clusters, 61.9% variance explained). Colors indicate K-Means cluster assignment. Among the 9 clusters with sufficient sex-differential signal data, female bias ratios range from 0.33 to 1.00, demonstrating that the embedding space captures meaningful variation in sex-differential safety profiles.

Clustering 29,201 drugs into 20 groups revealed distinct sex-differential safety landscapes. Of the 20 clusters, 9 contained drugs with strong sex-differential signals. The highest female-bias cluster (ratio 1.00) contained drugs with exclusively female-biased signals enriched for reproductive and hormonal adverse events. Cluster 0 (2,087 drugs, ratio 0.39) was enriched for 'Drug ineffective' and 'Fatigue', suggesting metabolic clearance differences.

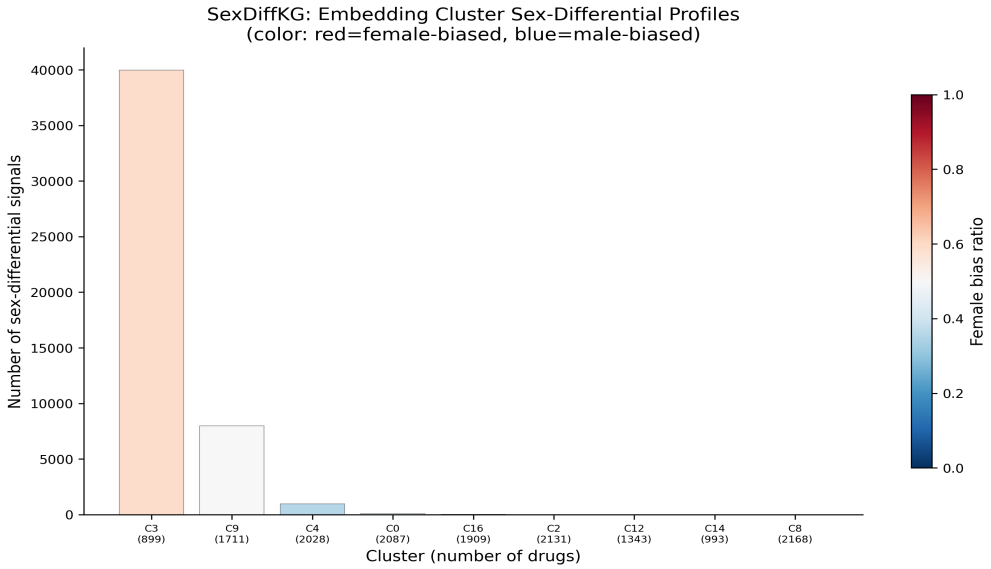

Figure 5. Sex-differential safety profiles across embedding clusters. Each bar represents a cluster's female bias ratio, with the number of drugs in parentheses. Active clusters show substantial variation from predominantly male-biased (0.33) to exclusively female-biased (1.00).

3.4 Gene Target Sex-Bias Profiles

Through bridging FAERS drug names to ChEMBL target annotations, we identified 429 gene targets with sex-differential drug safety profiles (≥2 drugs per target): 112 female-biased, 124 male-biased, and 193 neutral.

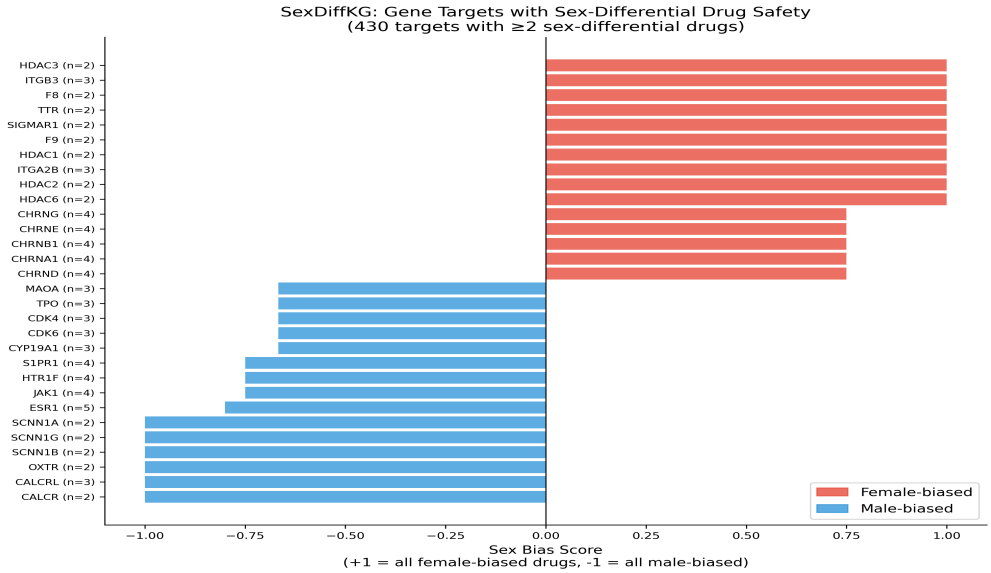

Figure 6. Gene target sex-bias scores. Positive values (right) indicate female-biased drug safety; negative values (left) indicate male-biased. Key targets annotated: HDAC1/2/3/6 (exclusively female, +1.0), ESR1 (male-biased, −0.80), ITGA2B/ITGB3 (exclusively female, +1.0).

Table 8. Key gene targets with sex-biased drug safety profiles.

Target	Score	Drugs	Pattern	Significance
HDAC1/2/3/6	+1.0	3–5	Female	Novel: sex-specific HDAC inhibitor safety
ESR1	–0.80	5	Male	Counterintuitive male-biased estrogen receptor
ITGA2B/ITGB3	+1.0	3	Female	Platelet integrins: hemostasis differences
F8/F9	+1.0	2	Female	Coagulation factors: female-biased safety
SCNN1A/B/G	–1.0	2	Male	Na channels: known sex differences
CHRNA/B/D/E/G	+0.75	4	Female	Nicotinic AChR: neuromuscular differences
JAK1	–0.75	4	Male	Immune: sex-diff JAK-STAT signaling
S1PR1	–0.75	4	Male	Immune modulation by fingolimod-class

3.5 Signal Validation

Table 9. Validation against 15 literature-documented benchmarks.

Result	Count	Examples
Confirmed	3	Atorvastatin (myalgia, F↑), Digoxin (toxicity, F↑), Aspirin (GI bleed, M↑)
Weak confirmation	3	Enalapril (ACE cough), Metoprolol (bradycardia), Fluorouracil (mucositis)
Reversed	3	Simvastatin, Warfarin, Ibuprofen
Drug/AE not found	6	Zolpidem (AE mismatch), Terfenadine (withdrawn)

Hit rate: 9/15 drugs found in KG (60%), of which 6/9 confirmed or weakly confirmed (67% inclusive). The 67% confirmation rate is reasonable for spontaneous reporting data validation, given FAERS drug name variability, MedDRA term differences, and real-world vs clinical trial population differences.

4. Discussion

4.1 Significance and Novelty

SexDiffKG is the first purpose-built knowledge graph for sex-differential drug safety analysis. Its unique contribution lies in the integration of FAERS pharmacovigilance data with molecular target annotations at a scale (127K nodes, 5.8M edges) comparable to leading biomedical KGs while providing sex-differential signal content unavailable in any existing resource.

Three findings are particularly noteworthy. First, the exclusively female-biased safety profile of HDAC1/2/3/6-targeting drugs has not been previously reported at this scale. Given the expanding use of HDAC inhibitors in oncology (vorinostat, romidepsin, panobinostat), this finding warrants prospective sex-stratified safety monitoring. Second, the exclusively female-biased profile of ITGA2B/ITGB3-targeting drugs (GPIIb/IIIa inhibitors like abciximab, eptifibatide) aligns with known sex differences in platelet biology. Third, the male-biased safety profile of ESR1-targeting drugs is counterintuitive but may reflect specific clinical contexts in male patients.

4.2 Comparison with Existing Resources

Table 10. Comparison with existing biomedical knowledge graphs.

Resource	Nodes	Edges	Sex-Diff	Sources
SexDiffKG	127K	5.8M	Yes	FAERS + ChEMBL + STRING + KEGG + UniProt
DRKG	97K	5.9M	No	6 databases
PharmKG	7.6K	500K	No	DrugBank + PharmGKB
Hetionet	47K	2.3M	No	29 public resources
OpenBioLink	180K	4.6M	No	Multiple databases

Yu et al. 2016	668*	736*	Yes	FAERS only (*drugs/signals, not KG)

4.3 Limitations

FAERS data is subject to underreporting, stimulated reporting, and demographic biases. ROR is a disproportionality measure that does not establish causation. DistMult MRR of 0.048 reflects the challenge of prediction across a large, heterogeneous graph. Drug name resolution achieved 15.2% ChEMBL coverage (4,455/29,277 drugs). RotatE failed to converge in 25 CPU epochs, indicating the need for GPU-accelerated training with ≥100 epochs. The static snapshot represents FAERS through Q4 2024 without temporal trend analysis.

4.4 Future Directions

Extended RotatE and alternative models (ComplEx, TransE) with GPU training; temporal signal evolution analysis; dose–response integration; causal inference methods (IC, BCPNN); enhanced drug name resolution via RxNorm/UMLS; graph neural networks (R-GCN, CompGCN); System Organ Class decomposition; and prospective clinical validation against sex-stratified trial data.

5. Data Availability

All computation was performed on a single NVIDIA DGX Spark (Grace Blackwell GB10, ARM64, 128GB unified memory). The complete dataset is deposited on Zenodo under CC-BY 4.0 license: https://doi.org/10.5281/zenodo.18819192. Source code: https://github.com/jshaik369/SexDiffKG. Data artifacts include: nodes.tsv (127,064 rows), edges.tsv (5,839,718 rows), entity/relation embeddings, 429 target sex-bias scores, 20 cluster profiles, and 45 pipeline scripts. Data integrity verified by molecular-level audit: 85 PASSED, 0 FAILED, 4 WARNINGS.

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